

**DATA266 Lab 1**

# **DATA266 Lab 1**

Team 5

Sneha Singh

Ritika Mukesh Neema

## Abstract

This report is Team 5's write-up for DATA266 Lab 1. Sneha Singh and Ritika Mukesh Neema each trained a character-level GPT on TinyStories, three Yelp polarity classifiers, and a CycleGAN for photo-to-Monet transfer. Each person used their own architecture and their own hyperparameters. Each part below gives one person's run, then the other person's run, then a comparison of the two. Kaggle FID scores are reported per member in Part 3.

The repository is [https://github.com/snehas-SJSU/Data\\_266-Lab1](https://github.com/snehas-SJSU/Data_266-Lab1).

## Introduction

Each part below has two individual write-ups and then one team comparison. Section A is Sneha's model, metrics, failures, and hardware. Section B is Ritika's. Section C puts the two runs side by side. Code for Sneha lives under `sneha_singh/`. Code for Ritika lives under `ritika_mukesh_neema/`. Raw TinyStories and the Monet and photo folders are linked from the README because they are too large for git. Yelp is downloaded by the notebook from Hugging Face.

## What each of us built

| Part                  | Sneha Singh (her own model)                                  | Ritika Mukesh Neema (her own model)                        |
|-----------------------|--------------------------------------------------------------|------------------------------------------------------------|
| 1 — GPT (TinyStories) | 4 layers, emb 256, context 128, 10 epochs, RTX 5090          | 4 layers, emb 128, context 256, 12 epochs, Colab Tesla T4  |
| 2 — Yelp polarity     | BiLSTM best, macro-F1 0.942, full train set, Colab Tesla T4  | BiLSTM best, macro-F1 0.934, full 560k set, Colab Tesla T4 |
| 3 — CycleGAN          | Batch 4, Colab Tesla T4, about 4.3 h, photo→Monet FID 105.06 | Batch 4, RTX 5090, about 28 min, FID 123.70                |

We share the raw data. The rows above are two separate runs of the same part, not a split of who had to do which part.

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## Part 1 — GPT from scratch

### 1.A Sneha Singh — individual

I trained a character-level GPT on TinyStories. The attention code is mine. I did not use `nn.Transformer` or `nn.MultiheadAttention`.

The tokenizer is a simple `char_to_idx / idx_to_char` map, not BPE. The model has 4 layers, 4 heads, embedding size 256, and a context of 128 characters. A causal mask stops each position from seeing the

future. I tied the token weights the usual way and trained with AdamW. Bias and LayerNorm do not get weight decay.

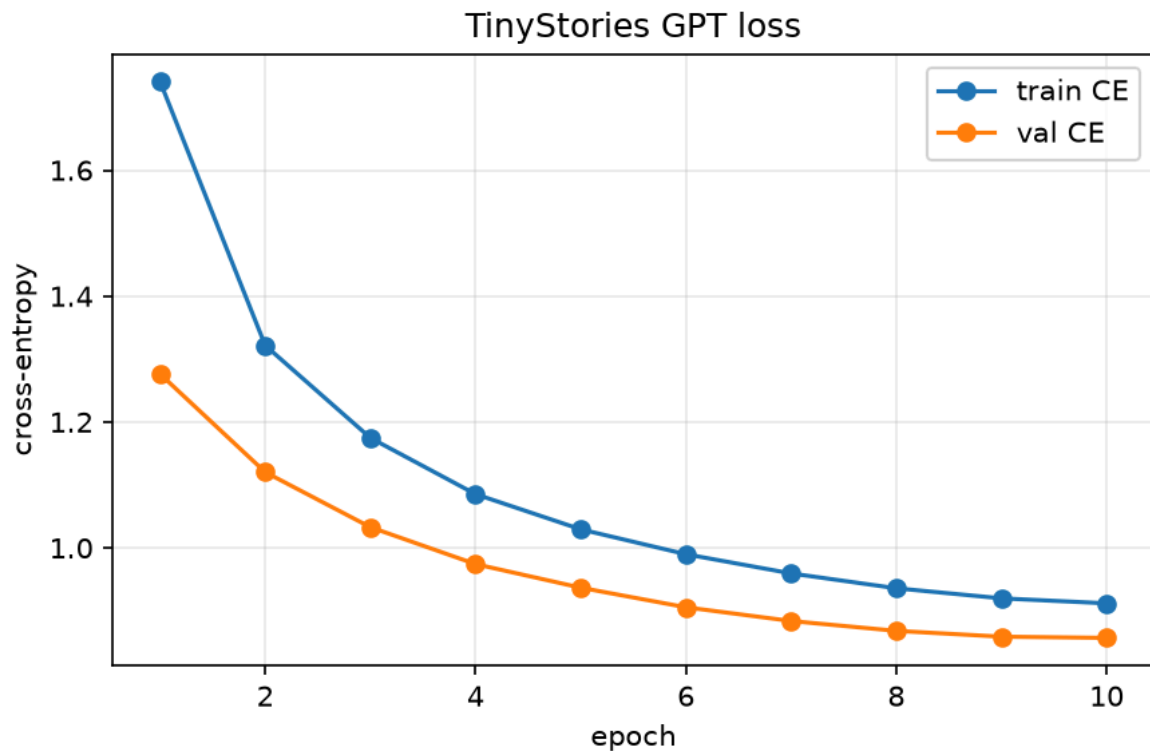
The reported run is the full one, not the old 2-epoch smoke test. I took my own split from TinyStories-train.txt: 100,000 train windows and 10,000 validation windows, seed 670. That text file is 1.79 GB, over GitHub's 100 MB limit, so it is not in git: [https://drive.google.com/drive/folders/12MFVOo6T3QRW3X6THiDkh5svtNgL13W?usp=share\\_link](https://drive.google.com/drive/folders/12MFVOo6T3QRW3X6THiDkh5svtNgL13W?usp=share_link)

I trained for 10 epochs on an NVIDIA GeForce RTX 5090 with mixed precision. Learning rate was 0.001, with 100 warmup steps and then cosine decay. Batch size was 32. The run took 457 seconds and peaked at about 349 MB. There were no NaN losses. The checkpoint best.pt is 12 MB, so it is on git.

Checkpoint id: task1\_llm/sneha\_singh/checkpoints/best.pt. Log: reproducibility/raw\_logs/sneha\_singh/task1\_llm/train\_full.log.

| Metric                           | Value                 |
|----------------------------------|-----------------------|
| Train cross-entropy              | 0.912                 |
| Val cross-entropy                | 0.857                 |
| Perplexity                       | 2.36                  |
| Bits per character               | 1.236                 |
| Generalization gap (val – train) | –0.055                |
| Top-1 next-character accuracy    | 0.729                 |
| Distinct-1 / 2 / 3               | 0.225 / 0.655 / 0.856 |
| Repeated 4-gram rate             | 0.120                 |
| Max gradient norm / NaN count    | 5.42 / 0              |
| Parameters                       | 3.25M                 |
| Train tokens/sec                 | 280,325               |
| Generation tokens/sec            | 409                   |
| Peak memory / train time         | 349 MB / 457 s        |

The result I care about on this model is that the samples stay varied while the loss stays stable. Distinct-1 is 0.225, distinct-2 is 0.655, and distinct-3 is 0.856. The repeated 4-gram rate is 0.120. Next-character top-1 accuracy is 0.729. Training had no NaNs, peaked at about 349 MB, and ran at about 280,000 tokens per second. The loss falls smoothly and the validation curve stays close to the training curve. That is what I wanted from 10 epochs on this size of model.



Part 1 training and validation loss

The prompt I used for samples was “some changes. She added some nice colors”.

### Failure cases

- **Repetition.** Greedy decoding loops: “You are very happy. You are very happy.”
- **Broken word.** The same decode prints “them them”. A character model does not know where a word ends.
- **Story drift.** Temperature 0.8 moves from colors to a box, a door, and a tree. The model only sees 128 characters at a time.

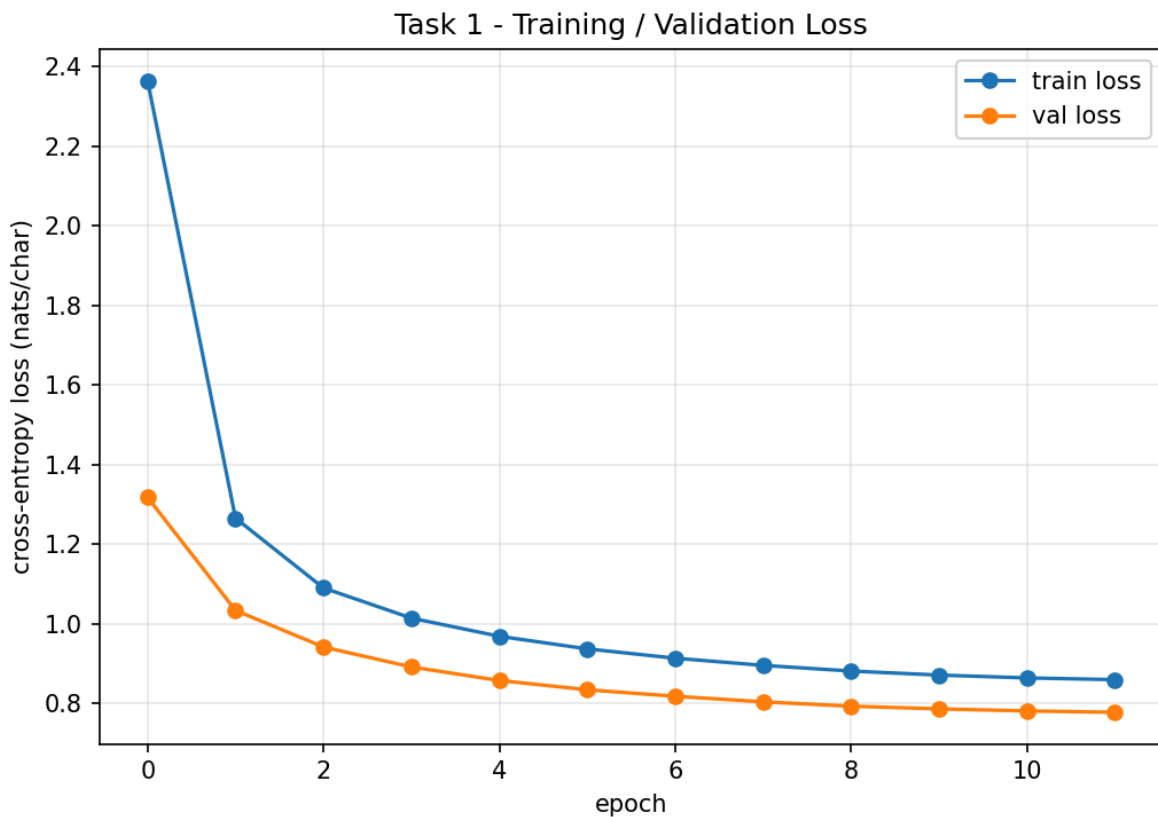
Full cases: [task1\\_llm/sneha\\_singh/failure\\_analysis.md](#)

## 1.B Ritika Mukesh Neema — individual

I wrote the attention myself. I did not use `nn.Transformer` or `nn.MultiheadAttention`. My model has 4 layers, 4 heads, embedding size 128, and a context of 256 characters. The tokenizer is character-level, built only from my training text, with a vocab of 221. I drew my own 100,000 / 10,000 windows from the full TinyStories train file, seed 6638. The checkpoint is `task1_llm/ritika_mukesh_neema/checkpoints/ckpt_final.pt`.

I trained for 12 epochs, batch 64, learning rate 0.0003, with linear warmup and cosine decay. Weight decay was 0.01 and gradients were clipped at 1.0. The run was on a Colab Tesla T4 and took 1,727 seconds, peaking at 1,749 MB. There were no NaN losses.

| Metric                        | Train | Val                   |
|-------------------------------|-------|-----------------------|
| Cross-entropy                 | 0.860 | 0.778                 |
| Perplexity                    | 2.36  | 2.18                  |
| Bits per character            | 1.240 | 1.122                 |
| Generalization gap            |       | -0.082                |
| Top-1 next-character accuracy |       | 0.755                 |
| Distinct-1 / 2 / 3            |       | 0.008 / 0.054 / 0.137 |
| Repeated 4-gram rate          |       | 0.206                 |
| Max gradient norm / NaN count |       | 4.99 / 0              |
| Parameters                    |       | 0.88M                 |
| Train tokens/sec              |       | 177,917               |
| Generation tokens/sec         |       | 239                   |
| Peak memory / train time      |       | 1,749 MB / 1,727 s    |



Part 1 training and validation loss

## Failure cases

- **Repetition.** All 10 greedy samples are the same story, and it loops: “You are very happy. You are very happy.”
- **Broken word.** Temperature sampling writes “designt”, which is not English.

- **Story drift.** The object moves from a box to a red ball to a bird.
- **Does not stop.** Generation does not stop at the story-boundary token, so a second story gets stuck on the end.

Full cases: [task1\\_llm/ritika\\_mukesh\\_neema/failure\\_analysis.md](#)

## 1.C Team comparison

|                            | Sneha                                                                | Ritika                                                               |
|----------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|
| Architecture               | 4 layers, 4 heads, emb 256, context 128, char tokenizer, causal mask | 4 layers, 4 heads, emb 128, context 256, char tokenizer, causal mask |
| Hyperparameters            | 10 epochs, batch 32, lr 0.001, warmup 100, seed 670, AdamW           | 12 epochs, batch 64, lr 0.0003, warmup + cosine, seed 6638, clip 1.0 |
| Train / val CE             | 0.912 / 0.857                                                        | 0.860 / 0.778                                                        |
| Perplexity / bits per char | 2.36 / 1.236                                                         | 2.18 / 1.122                                                         |
| Gen. gap / top-1 acc       | -0.055 / 0.729                                                       | -0.082 / 0.755                                                       |
| Distinct-1/2/3             | 0.225 / 0.655 / 0.856                                                | 0.008 / 0.054 / 0.137                                                |
| Repeated 4-gram / NaNs     | 0.120 / 0                                                            | 0.206 / 0                                                            |
| Max gradient norm          | 5.42                                                                 | 4.99                                                                 |
| Params / time / peak MB    | 3.25M / 457 s / 349                                                  | 0.88M / 1,727 s / 1,749                                              |
| Train / gen tokens per sec | 280,325 / 409                                                        | 177,917 / 239                                                        |
| Checkpoint                 | best.pt                                                              | ckpt_final.pt                                                        |
| Log                        | train_full.log                                                       | train_full.log                                                       |

**Strength.** Ritika’s validation loss is 0.778, and Sneha’s is 0.857. Sneha’s distinct-1 is 0.225, and Ritika’s is 0.008. Each run leads on a different metric.

**Weakness.** Both greedy decoders loop on “You are very happy.”

**Limitation.** Embedding size, context length, and the machine differ, so the loss gap is not a pure architecture contest.

**Next.** Both of us would add a repetition penalty at decode time, and stop generation when the story-boundary token appears.

## Part 2 — Yelp polarity

### 2.A Sneha Singh — individual

This is Yelp polarity, not IMDB. I learned the embeddings from scratch. I did not use a pretrained language model.

The notebook downloads fancyzhx/yelp\_polarity from Hugging Face when the run starts.

I wanted one simple model and two that can use word order, so a strong baseline would be obvious if the fancier models did not beat it.

1. **Baseline.** Mean pool over the embeddings, then a linear layer.
2. **BiLSTM.** Reads the review in both directions, so negation and longer sentences have a chance.
3. **TextCNN.** Filters of width 3, 4, and 5, meant to catch short phrases like “not good”.

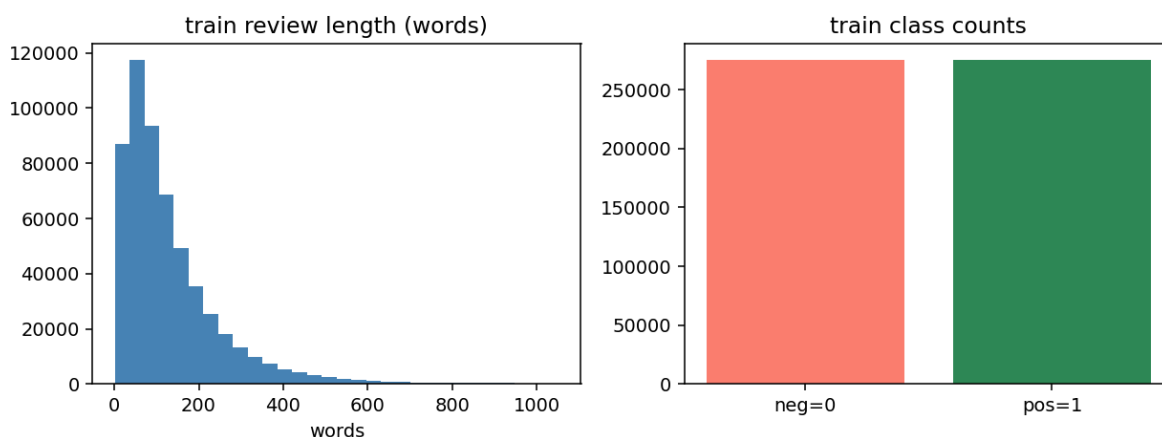
I lowercased, stripped punctuation, removed stopwords, and lemmatized. The vocabulary comes from the training text only. I held out 10,000 validation reviews from the 560,000-row train file, trained on 549,953 reviews, and tested on all 38,000 official test reviews. Five epochs, batch 64, embedding size 100, maximum length 128. All three models ran on a Colab Tesla T4. Peak memory was 228 MB, 344 MB, and 1,656 MB.

Checkpoints: task2\_sentiment/sneha\_singh/checkpoints/baseline\_full.pt, experimental\_a\_full.pt, and experimental\_b\_full.pt. Logs: reproducibility/raw\_logs/sneha\_singh/task2\_sentiment/baseline\_full.log, experimental\_a\_full.log, and experimental\_b\_full.log.

| Model    | Acc   | Macro-F1 | ROC-AUC | MCC   | Brier | ECE   | Time   |
|----------|-------|----------|---------|-------|-------|-------|--------|
| Baseline | 0.932 | 0.932    | 0.979   | 0.863 | 0.052 | 0.006 | 233 s  |
| BiLSTM   | 0.942 | 0.942    | 0.986   | 0.883 | 0.044 | 0.012 | 794 s  |
| TextCNN  | 0.937 | 0.937    | 0.984   | 0.875 | 0.047 | 0.008 | 3646 s |

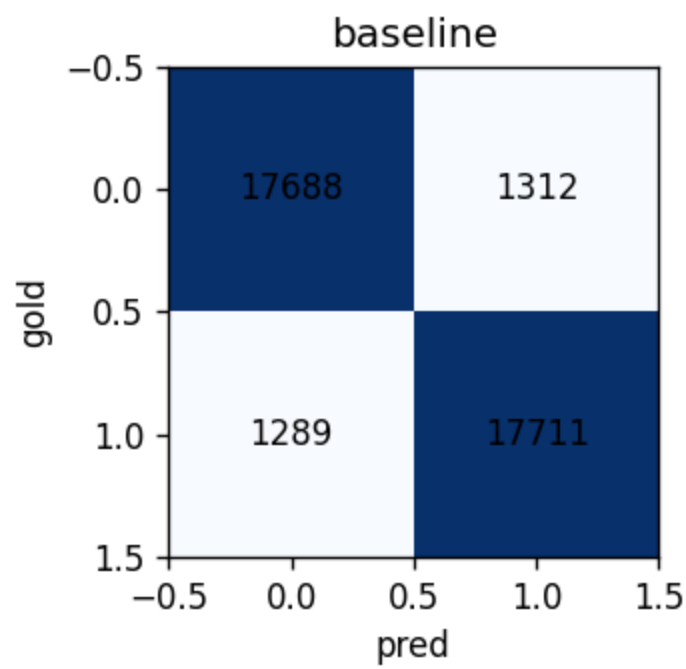
The BiLSTM has the best macro-F1. McNemar against the baseline is  $p < 0.001$  for both the BiLSTM and the TextCNN, so on this 38k test set both beat the mean pool. Short reviews are a bit easier than long ones. Long-review macro-F1 is 0.927, 0.934, and 0.933 for baseline, BiLSTM, and TextCNN.

The length and class balance of this run:

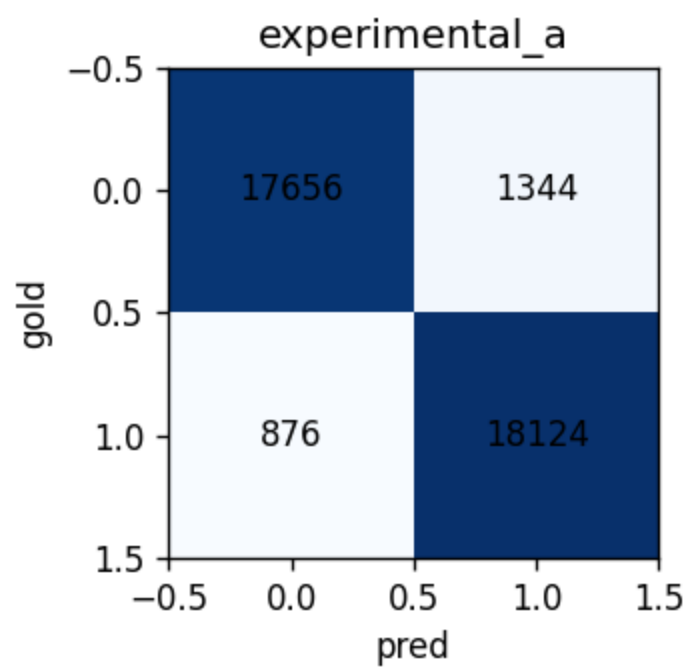


Yelp length and class balance

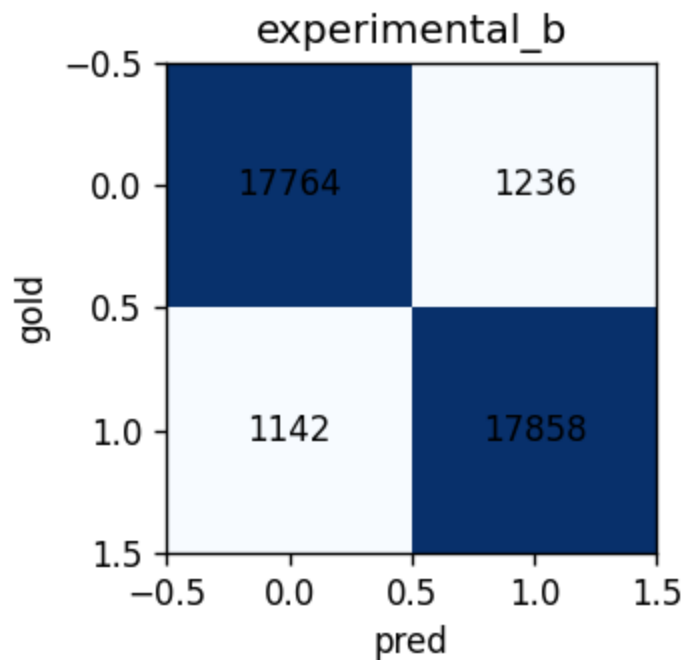
Confusion matrices on the test set:



Baseline confusion matrix



BiLSTM confusion matrix



TextCNN confusion matrix

## Error review

I read 20 BiLSTM mistakes: five confident false positives, five confident false negatives, five near the decision threshold, and five long-review misses.

- **Confident false positive.** Gold negative, predicted positive: “wow love place everything clean new”.
- **Confident false negative.** Gold positive, predicted negative: “look know cox suck fact terrible business”.
- **Near the threshold.** “wow seems like taco bell arizona”.
- **Long review.** “given mixed review quite sure expect”.

The ones that show up most are praise words inside a negative review, a complaint that later turns around, and long reviews where the stars and the sentences pull apart. The fix I wrote down for these cases is to balance review length or train longer, and to add negation handling.

Full 20-row table: [task2\\_sentiment/sneha\\_singh/failure\\_analysis.md](https://github.com/sneha-singh/failure-analysis.md)

## 2.B Ritika Mukesh Neema — individual

I trained the same three families, with embeddings learned from scratch. My baseline is a mean pool plus a linear layer. My BiLSTM is there so negation can depend on word order. My TextCNN uses filter widths 3, 4, and 5 with global max-pool. I lowercased, stripped punctuation and HTML, removed stopwords, and used Porter stemming instead of lemmatization. I trained on the full Yelp polarity set, about 560,000 reviews, on a Colab Tesla T4. Checkpoints are

task2\_sentiment/ritika\_mukesh\_neema/checkpoints/baseline.pt, bilstm.pt, and textcnn.pt. I do not have confusion-matrix images in the repo. The numbers below are from my metrics\_report.csv.

| Model    | Acc   | Macro-F1 | ROC-AUC | PR-AUC | MCC   | Brier | ECE   | Time  | Peak MB |
|----------|-------|----------|---------|--------|-------|-------|-------|-------|---------|
| Baseline | 0.922 | 0.922    | 0.974   | 0.973  | 0.843 | 0.059 | 0.007 | 125 s | 2,123   |
| BiLSTM   | 0.934 | 0.934    | 0.982   | 0.982  | 0.868 | 0.051 | 0.025 | 798 s | 2,274   |
| TextCNN  | 0.926 | 0.926    | 0.979   | 0.979  | 0.852 | 0.055 | 0.021 | 267 s | 2,450   |

My BiLSTM is the best of my three. McNemar against my baseline is  $p < 0.001$  for the BiLSTM and  $p = 0.00022$  for the TextCNN, so on my test set both sequence models beat the mean pool.

## Error review

I reviewed 20 TextCNN mistakes.

- **Confident false positive.** Sarcasm, with no negative words: “Is there REALLY even a Leonard...”  
A French review in the same set also fails, because the stemmer is English-only.
- **Confident false negative.** A positive review opens with “Cox sucks” before it turns around.
- **Near the threshold.** “The employees at this Target seemed unusually friendly...”
- **Long review.** “I used to love D&B... it has gone down hill.”

The fixes I wrote down are a language filter for non-English reviews, more weight on the words after “however”, and treating a score near 0.5 as a calibration margin rather than a hard error.

Full 20-case write-up: [task2\\_sentiment/ritika\\_mukesh\\_neema/failure\\_analysis.md](#)

## 2.C Team comparison

Same three model families. The tables use the same columns. Both of us trained on the full Yelp train file. Sneha tested on the official 38,000 reviews. Ritika tested on her own split of that file.

**Sneha Singh.** 549,953 / 10,000 / 38,000, lemmatize, max length 128, Colab Tesla T4. Checkpoints: baseline\_full.pt, experimental\_a\_full.pt, experimental\_b\_full.pt.

| Model    | Acc   | Macro-F1 | ROC-AUC | PR-AUC | MCC   | Brier | ECE   | McNemar p | Params | Time   | Peak MB |
|----------|-------|----------|---------|--------|-------|-------|-------|-----------|--------|--------|---------|
| Baseline | 0.932 | 0.932    | 0.979   | 0.979  | 0.863 | 0.052 | 0.006 | —         | 10.68M | 233 s  | 228     |
| BiLSTM   | 0.942 | 0.942    | 0.986   | 0.986  | 0.883 | 0.044 | 0.012 | < 0.001   | 10.76M | 794 s  | 344     |
| TextCNN  | 0.937 | 0.937    | 0.984   | 0.985  | 0.875 | 0.047 | 0.008 | < 0.001   | 10.75M | 3646 s | 1,656   |

**Ritika Mukesh Neema.** Full 560k set, Porter stem, max length 200, Colab Tesla T4. Checkpoints: baseline.pt, bilstm.pt, textcnn.pt.

| Model    | Acc   | Macro-F1 | ROC-AUC | PR-AUC | MCC   | Brier | ECE   | McNemar p | Params | Time  | Peak MB |
|----------|-------|----------|---------|--------|-------|-------|-------|-----------|--------|-------|---------|
| Baseline | 0.922 | 0.922    | 0.974   | 0.973  | 0.843 | 0.059 | 0.007 | —         | 3.00M  | 125 s | 2,123   |
| BiLSTM   | 0.934 | 0.934    | 0.982   | 0.982  | 0.868 | 0.051 | 0.025 | < 0.001   | 3.24M  | 798 s | 2,274   |
| TextCNN  | 0.926 | 0.926    | 0.979   | 0.979  | 0.852 | 0.055 | 0.021 | 0.00022   | 3.12M  | 267 s | 2,450   |

Precision, recall, the other F1 scores, bootstrap intervals, examples per second, and slice scores are in the next table. Slice scores are filled where the metrics file has a short-review or long-review column.

| Metric           | Sneha baseline | Sneha BiLSTM | Sneha TextCNN | Ritika baseline | Ritika BiLSTM | Ritika TextCNN |
|------------------|----------------|--------------|---------------|-----------------|---------------|----------------|
| Precision macro  | 0.932          | 0.942        | 0.937         | 0.922           | 0.934         | 0.926          |
| Recall macro     | 0.932          | 0.942        | 0.937         | 0.922           | 0.934         | 0.926          |
| F1 micro         | 0.932          | 0.942        | 0.937         | 0.922           | 0.934         | 0.926          |
| F1 weighted      | 0.932          | 0.942        | 0.937         | 0.922           | 0.934         | 0.926          |
| Acc 95% CI       | 0.929–0.934    | 0.939–0.944  | 0.935–0.940   | 0.919–0.925     | 0.931–0.936   | 0.924–0.928    |
| Macro-F1 95% CI  | 0.929–0.934    | 0.939–0.944  | 0.935–0.940   | 0.919–0.925     | 0.931–0.936   | 0.923–0.928    |
| MCC 95% CI       | 0.859–0.868    | 0.879–0.888  | 0.870–0.879   | 0.838–0.849     | 0.863–0.873   | 0.847–0.857    |
| Examples/sec     | 11,807         | 3,464        | 754           | 20,181          | 3,159         | 9,432          |
| Short macro-F1   | 0.934          | 0.947        | 0.940         |                 |               |                |
| Short error rate | 0.065          | 0.052        | 0.059         |                 |               |                |
| Long macro-F1    | 0.927          | 0.934        | 0.933         |                 |               |                |
| Long error rate  | 0.072          | 0.065        | 0.066         |                 |               |                |

**Strength.** Sneha’s BiLSTM macro-F1 is 0.942. Ritika’s BiLSTM macro-F1 is 0.934. On both runs the BiLSTM beats that person’s baseline, and McNemar says the lead is real.

**Weakness.** Both of us miss sarcasm, mixed reviews, and negation.

**Limitation.** The test splits are not the same cut of the file. Sneha lemmatizes and uses a maximum length of 128. Ritika stems and uses a maximum length of 200. The scores are not one shared test.

**Next.** Both of us would use the same maximum review length, so long reviews are scored the same way.

## Part 3 — CycleGAN, photo to Monet

### 3.A Sneha Singh — individual

I trained an unpaired CycleGAN. The Kaggle direction is photo to Monet. The generators are mine. I did not use a pretrained model to make the images. Inception and the other nets in the eval script are only for measuring.

Each generator is a ResNet with 9 blocks at  $256 \times 256$ , reflection padding, instance norm, and a tanh output. Upsampling is nearest-neighbor times two, then a stride-1 convolution, not a transposed convolution. Each discriminator is a PatchGAN trained with least-squares loss. The losses are adversarial, cycle L1 with  $\lambda = 10$ , and identity at half of that. Real labels are smoothed to 0.9. An image pool of 50 feeds the discriminator. Batch size is 4. I trained 40 epochs at a constant learning rate and 40 more with decay, and each epoch resamples about 800 photos. The shorter Monet loader is cycled so those photos actually get used. pred\_B2A is photo→Monet (7,038 images). pred\_A2B is Monet→photo (300 images). Those are the professor folder names.

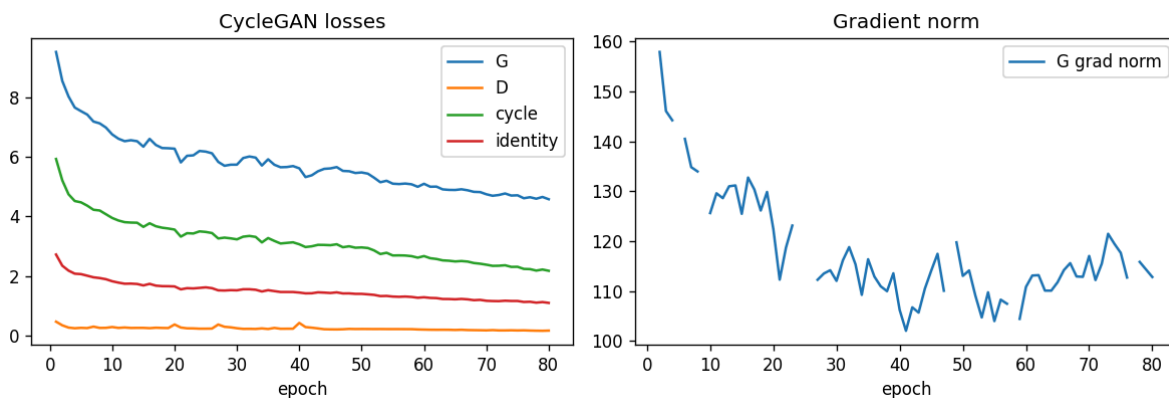
Training was on a Colab Tesla T4 with mixed precision. It took about 4.3 hours and peaked near 6568 MB. The log line is device=cuda. best.pt is 107.9 MB, over GitHub's 100 MB limit, so it is not in git: [https://drive.google.com/drive/folders/12AgM95RbZAQUyouH7sukM9nu\\_rVTD3C9?usp=share\\_link](https://drive.google.com/drive/folders/12AgM95RbZAQUyouH7sukM9nu_rVTD3C9?usp=share_link)

The Monet and photo folders are in one Drive folder: [https://drive.google.com/drive/folders/1BXYfhW8uZ6umK1TZFW8Un62mZVK72L7Y?usp=share\\_link](https://drive.google.com/drive/folders/1BXYfhW8uZ6umK1TZFW8Un62mZVK72L7Y?usp=share_link)

| Direction                | FID    | KID   | MiFID | LPIPS | Content cosine | Cycle L1 |
|--------------------------|--------|-------|-------|-------|----------------|----------|
| Photo → Monet (pred_B2A) | 105.06 | 0.018 | 0.405 | 0.452 | 0.532          | 0.112    |
| Monet → photo (pred_A2B) | 109.33 | 0.030 | 0.425 | 0.365 | 0.586          | 0.103    |

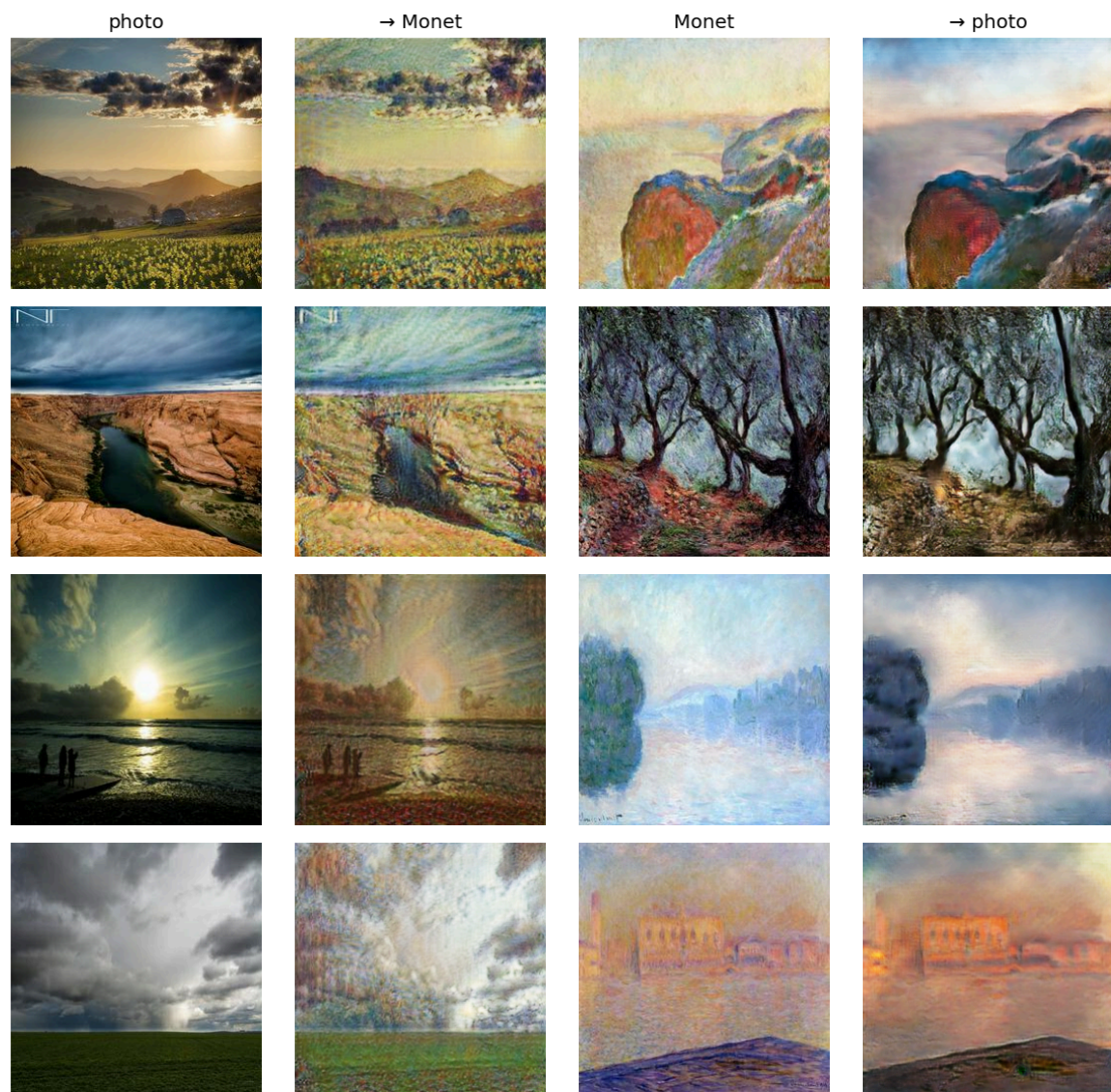
FID is the Fréchet distance between Inception features of the generated images and the real target images. MiFID is the average cosine distance of those features on the first 300 sorted images, which is the professor script. Lower is better on both. submission.csv is the average of both directions: FID 107.20, MiFID 0.415.

Losses fell across the 80 epochs. The discriminator loss ended near 0.17, which is low. The generator is not winning that fight, and I think that is why the brush texture gets noisy.



CycleGAN losses and gradient norm

The grid is four photos, the Monet version of each, four real Monets, and the photo version of those.



Photo, generated Monet, real Monet, generated photo

## Human audit

What I see in that grid, and in the 30 images I audited:

- **Grain.** Skies and water pick up a repeated dab texture.
- **Color wash.** Some skies go flat, and sunsets brown out at the edges.
- **Soft reverse.** A few Monet-to-photo frames go soft or pick up a dark blob.
- **Content.** The layout of the photo usually survives. I did not mark any of my 30 as a copied training Monet.

On the 0–2 sheet my averages were style 1.4 and content 1.7, where higher is better, and artifacts 1.4, where higher is worse.

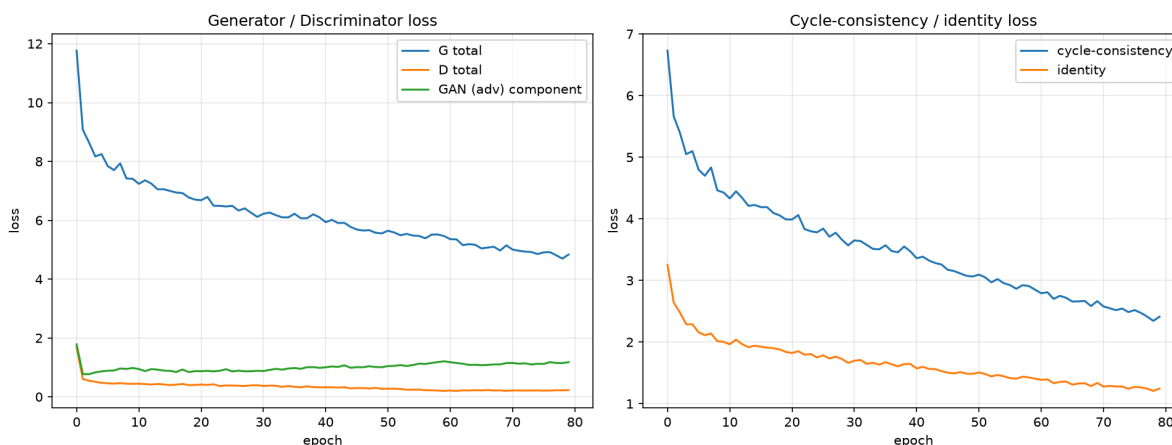
Full notes: [task3\\_gan/sneha\\_singh/failure\\_analysis.md](https://task3_gan/sneha_singh/failure_analysis.md)

### 3.B Ritika Mukesh Neema — individual

I trained my own CycleGAN: two ResNet generators with 9 blocks, instance norm, and reflection padding, plus two 70×70 PatchGAN discriminators. I used least-squares adversarial loss, cycle L1 with  $\lambda = 10$ , identity loss with  $\lambda = 5$ , and an image pool of 50. Batch size was 4. I trained 40 epochs at a constant learning rate and 40 with decay, on an NVIDIA RTX 5090. The run took 1,660 seconds and peaked at 1,645 MB. There were no NaN losses. My photo-to-Monet direction is the one I label B2A. My photo-to-Monet FID is 123.70 (local, against my own training-data split). `ckpt_final.pt` is 107.9 MB, so it is not in git: <https://drive.google.com/file/d/1Wlu22EKbp4QATijRbNBbDOFFGHLHUiNj/view>. My Kaggle submission (`kaggle_submission.csv`, computed against the competition's official `real_stats.npz` with the same Inception feature extractor as `evaluate_local.py`) scored FID 104.30, MiFID 0.413 — public leaderboard score -52.3574, submitted under my own name.

| Direction              | FID    | KID   | Precision | Recall | Cycle L1 | LPIPS | Content cosine |
|------------------------|--------|-------|-----------|--------|----------|-------|----------------|
| Photo → Monet (my B2A) | 123.70 | 0.027 | 0.327     | 0.570  | 0.126    | 0.337 | 0.775          |
| Monet → photo (my A2B) | 120.72 | 0.040 | 0.613     | 0.207  | 0.109    | 0.405 | 0.873          |

Final generator loss was 4.84, discriminator loss 0.23, cycle loss 2.41, identity loss 1.24. Mean gradient norm was 57, max 315, NaN count 0. About 28.3 million parameters, 14.5 images/sec.



#### CycleGAN losses

My `submission.csv` now holds the officially-scored Kaggle submission: FID 104.30, MiFID 0.413, public score -52.3574. The 123.70 FID above is a local-only number against my own training-data split, not the competition's official `real_stats.npz`.

#### Failure notes

- **Cycle check.** Cycle L1, LPIPS, and content cosine are on 100 images, not the full photo set.
- **Human audit.** See `task3_gan/ritika_mukesh_neema/failure_analysis.md` for the full write-up. The visual human-audit pass (`src/human_audit.py`, 30 blinded samples) is specified but not yet

run.

### 3.C Team comparison

Both models are ResNet-9 CycleGANs with a PatchGAN and least-squares loss. Lower FID is better. Checkpoint links are in sections 3.A and 3.B.

|                       | Sneha photo → Monet | Sneha Monet → photo | Ritika photo → Monet | Ritika Monet → photo |
|-----------------------|---------------------|---------------------|----------------------|----------------------|
| Epochs / batch        | 40 + 40, batch 4    | same run            | 40 + 40, batch 4     | same run             |
| FID                   | 105.06              | 109.33              | 123.70               | 120.72               |
| KID                   | 0.018               | 0.030               | 0.027                | 0.040                |
| Precision             | 0.540               | 0.660               | 0.327                | 0.613                |
| Recall                | 0.293               | 0.089               | 0.570                | 0.207                |
| Cycle L1              | 0.112               | 0.103               | 0.126                | 0.109                |
| LPIPS                 | 0.452               | 0.365               | 0.337                | 0.405                |
| Content cosine        | 0.532               | 0.586               | 0.775                | 0.873                |
| Params                | 28.3M               | same run            | 28.3M                | same run             |
| Time / images per sec | 15,342 s / 8.34     | same run            | 1,660 s / 14.5       | same run             |
| Peak memory           | 6,568 MB            | same run            | 1,645 MB             | same run             |
| NaN count             | 0                   | same run            | 0                    | same run             |
| Generator loss        | 4.574               | same run            | 4.835                | same run             |
| Discriminator loss    | 0.168               | same run            | 0.227                | same run             |
| Cycle loss            | 2.174               | same run            | 2.412                | same run             |
| Identity loss         | 1.097               | same run            | 1.244                | same run             |
| Gradient norm         | 112.82              | same run            | 57.00 mean, 315 max  | same run             |

**Human audit score** on the 30 photo → Monet images. Style and content are 0–2, higher is better. Artifacts are 0–2, higher is worse. Cohen’s kappa is the inter-rater agreement for both of us on the same 30 images.

|           | Sneha | Ritika | Cohen’s kappa |
|-----------|-------|--------|---------------|
| Style     | 1.4   |        |               |
| Content   | 1.7   |        |               |
| Artifacts | 1.4   |        |               |

**Strength.** Sneha’s photo-to-Monet FID is 105.06. Ritika’s FID is 123.70. Ritika’s run is the faster one, about 28 minutes against about 4.3 hours.

**Weakness.** Both discriminator losses ended low, 0.17 for Sneha and 0.23 for Ritika. On Sneha’s images the brush texture is grainy.

**Limitation.** Batch size, training time, and the number of images in the cycle scores differ, so the two FID numbers are not a pure architecture contest.

**Next.** Later full-photo retraining scored worse, so the class upload stays this checkpoint: `submission.csv` with the average FID and MiFID of both directions.

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## Final comparison

| Part                  | Sneha Singh                                                  | Ritika Mukesh Neema                            |
|-----------------------|--------------------------------------------------------------|------------------------------------------------|
| 1 — validation loss   | 4 layers, emb 256, context 128. 0.857                        | 4 layers, emb 128, context 256. 0.778          |
| 1 — distinct-1        | 4 layers, emb 256, context 128. 0.225                        | 4 layers, emb 128, context 256. 0.008          |
| 2 — best model        | BiLSTM (lemmatize), 550k train, 38k test. Macro-F1 0.942     | BiLSTM (Porter stem), full set. Macro-F1 0.934 |
| 3 — photo → Monet FID | Batch 4, about 4.3 h. 105.06 (submission average FID 107.20) | Batch 4, about 28 min. 123.70                  |

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## Kaggle score

Team `PairProgramming_Team_05`. Score uses the new professor evaluation script:  $(\text{FID} + \text{MiFID}) / 2$ , each averaged over both directions. Current `submission.csv`: FID 107.20, MiFID 0.415.

| Team                                 | Public score                                       | Rank                    |
|--------------------------------------|----------------------------------------------------|-------------------------|
| <code>PairProgramming_Team_05</code> | $\approx 53.81$ ( <i>fill exact Kaggle value</i> ) | <i>fill from Kaggle</i> |

## Citation

1. Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention Is All You Need. 2017. <https://arxiv.org/abs/1706.03762>
2. Ronen Eldan and Yuanzhi Li. TinyStories: How Small Can Language Models Be and Still Speak Coherent English? 2023. <https://arxiv.org/abs/2305.07759>
3. Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros. Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. 2017. <https://arxiv.org/abs/1703.10593>